



NORMANHURST BOYS HIGH SCHOOL

MATHEMATICS EXTENSION 2

2020 Year 12 Course Assessment Task 1

Monday December 2, 2019

General instructions

- Working time – 1 hour.
(plus 5 minutes reading time)
- Write your answers in the space provided.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- NESA approved calculators may be used.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.
- Attempt **all** questions.
- At the conclusion of the examination, bundle any additional writing booklets used in the correct order within this paper and hand to examination supervisors.
- A NESA Reference Sheet is provided.

Class (please ✓)

- ☐ 12MAT.1 – Ms Ham
- ☐ 12MAT.2 – Mr Lam

NESA STUDENT #: # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	6	Total	%
MARKS	$\overline{6}$	$\overline{5}$	$\overline{7}$	$\overline{4}$	$\overline{12}$	$\overline{5}$	$\overline{39}$	

Glossary

- $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3\}$ – set of all integers.
- $\mathbb{Z}^+$ – all positive integers (excludes zero)
- $\mathbb{R}$ – set of all real numbers

Question 1 (6 Marks)**Marks**(a) If $z = 1 + 2i$, find in the form $a + ib$, where $a, b \in \mathbb{R}$, the values of:i. z^2 **1**

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ii. $\frac{1}{z}$ **1**

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(b) i. Express $z = -1 + \sqrt{3}i$ in the form $z = re^{i\theta}$. **2**

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ii. Hence show that $z^8 - 16z^4$ is purely imaginary. **2**

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Question 2 (5 Marks)**Marks**

- (a) i. Find the values of the real numbers a and b such that $(a + ib)^2 = 5 - 12i$. **2**

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- ii. Hence or otherwise, solve the equation $ix^2 + 3x + (3 - i) = 0$. **2**

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- (b) The complex number z has $|z| = 1$ and $\text{Arg}(z) = \frac{\pi}{6}$. The points represented by the complex numbers z , 0 and ω form an equilateral triangle on the Argand diagram. **1**

Find a possible complex number ω .

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Question 3 (7 Marks)**Marks**

A circle C and ray ℓ have equations $|z - 2\sqrt{3} - i| = 4$ and $\text{Arg}(z + i) = \frac{\pi}{6}$ respectively.

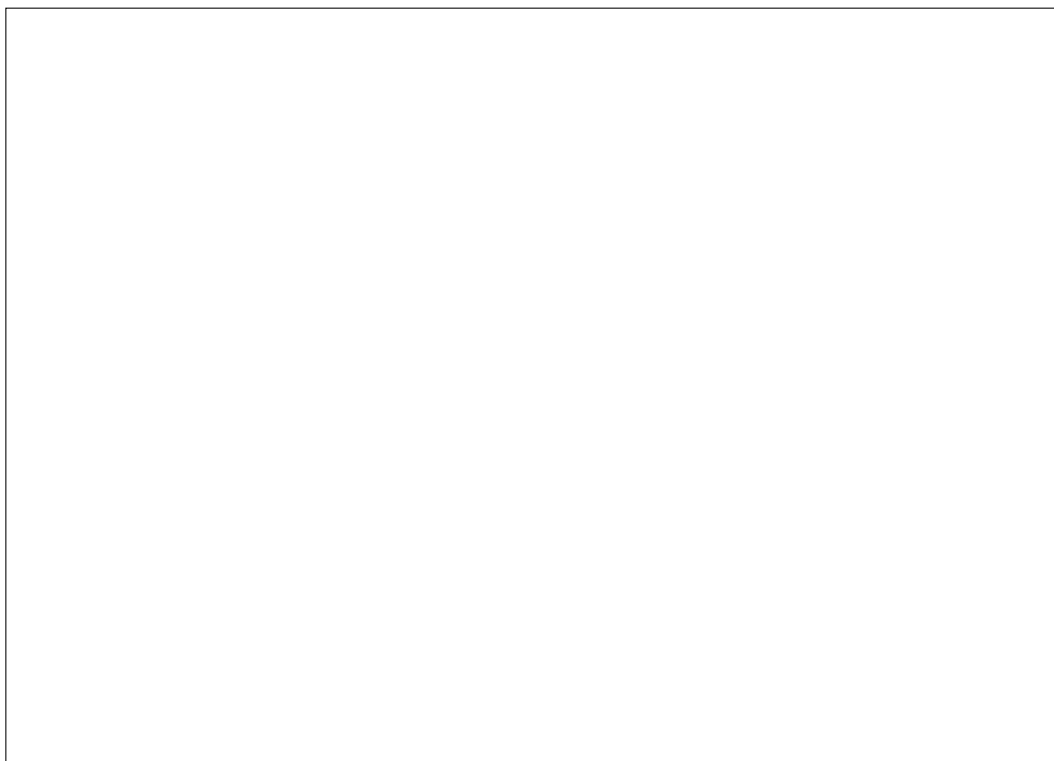
- (a) Show that the circle C passes through the point where $z = -i$. **1**

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- (b) Show that the ray ℓ passes through the centre of C . **2**

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- (c) Sketch C and ℓ on the same Argand diagram. **2**



- (d) Shade on your sketch in part (c), the region satisfying both **2**

$$|z - 2\sqrt{3} - i| \leq 4 \quad \text{and} \quad 0 \leq \text{Arg}(z + i) \leq \frac{\pi}{6}$$

Examination continues overleaf...

Question 4 (4 Marks)**Marks**

In the Argand diagram, $OABC$ is a square, where O , A , B and C are in anti-clockwise cyclic order. The complex number z is represented by the vector $\overrightarrow{OA}$.

- (a) Find in terms of z , the complex numbers represented by the vectors $\overrightarrow{OC}$ and $\overrightarrow{OB}$. **2**

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- (b) If the vector $\overrightarrow{OB}$ represents the complex number $4 + 2i$, find z in the form $a + ib$, where a and $b \in \mathbb{R}$. **2**

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Examination continues overleaf...

Question 5 (12 Marks)**Marks**

- (a) The cubic equation

$$2z^3 + pz^2 + qz + 16 = 0$$

where $p, q \in \mathbb{R}$, has roots α, β and γ .

It is given that $\alpha = 2 + 2\sqrt{3}i$.

- i. Write down another root, β , of the equation.

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- ii. Find the third root γ .

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- iii. Find the values of p and q .

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- iv. By expressing α in modulus-argument form, show that

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$$\left(2 + 2\sqrt{3}i\right)^n = 4^n \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3}\right)$$

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Question 5 continues overleaf...

Question 5 continued from the previous page...

- v. Hence, show that

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$$\alpha^n + \beta^n + \gamma^n = 2^{2n+1} \cos \frac{n\pi}{3} + \left(-\frac{1}{2}\right)^n$$

where $n \in \mathbb{Z}$.

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- (b) i. If $z = e^{i\theta}$, show that

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$$z + z^{-1} = 2 \cos \theta$$

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- ii. Given further that $z + z^{-1} = \sqrt{2}$, find the value of

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$$z^{10} + \frac{1}{z^{10}}$$

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Examination continues overleaf...

Question 6 (5 Marks)**Marks**

- (a) i. Solve $z^4 + 1 = 0$, giving your answers in modulus-argument form. **3**

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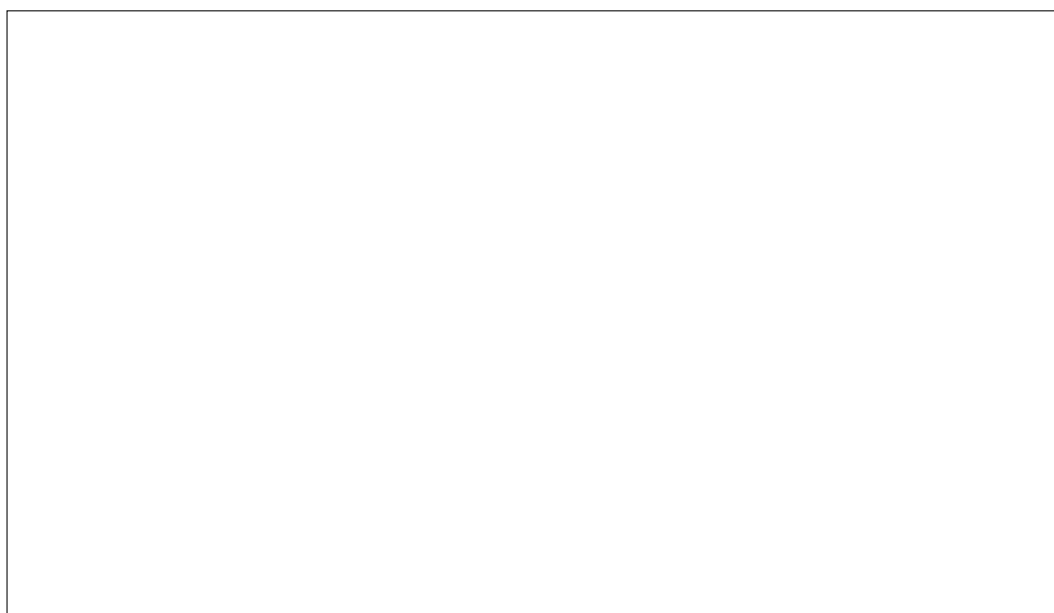
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- ii. Plot these solutions on the Argand diagram. **1**



- iii. Find the exact area of the quadrilateral that they form. **1**

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End of paper.

2020 Mathematics Extension 2 Year 12 Course Assessment Task 1 STUDENT SELF REFLECTION

1. In hindsight, did I do the best I can? Why or why not?

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- Q4 - Complex Numbers (Vectors)

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- Q4 - Complex Numbers (De Moivre's Theorem/Polynomials)

2. Which topics did I need more help with, and what parts specifically?

- Q1 - Complex Numbers (Arithmetic)

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- Q5 - Complex Numbers (Roots of unit/polynomials)

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- Q2 - Complex Numbers (Equations)

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3. What other parts from the feedback session can I take away to refine my solutions for future reference?

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- Q3 - Complex Numbers (Graphs in the Argand Diagram)

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Sample Band E4 Responses

Question 1 (Ham)

- (a) i. (1 mark)

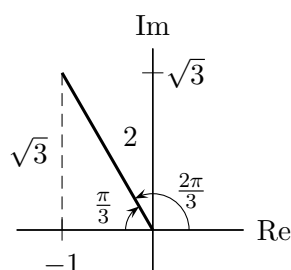
$$\begin{aligned}(1+2i)^2 &= (1^2 - 2^2) + 4i \\ &= -3 + 4i\end{aligned}$$

- ii. (1 mark)

$$\begin{aligned}\frac{1}{1+2i} \times \frac{1-2i}{1-2i} &= \frac{1-2i}{1^2 + 2^2} \\ &= \frac{1}{5} - \frac{2}{5}i\end{aligned}$$

- (b) i. (2 marks)

- ✓ [1] for modulus
✓ [1] for argument



$$z = -1 + \sqrt{3}i = 2e^{i(\frac{2\pi}{3})}$$

- ii. (2 marks)

- ✓ [1] for finding the correct value of z^4 .
✓ [1] for showing $z^8 - 16z^4 \notin \mathbb{R}$.

$$\begin{aligned}z^4 &= \left(2e^{i(\frac{2\pi}{3})}\right)^4 \\ &= 2^4 e^{i(\frac{8\pi}{3})} \\ &= 16e^{i(\frac{2\pi}{3})} \\ z^8 - 16z^4 &= z^4 (z^4 - 16) \\ &= 16e^{i(\frac{2\pi}{3})} \left(16e^{i(\frac{2\pi}{3})} - 16\right) \\ &= 256 \left(e^{i(\frac{4\pi}{3})} - e^{i(\frac{2\pi}{3})}\right) \\ &= 256 \left(e^{i(-\frac{2\pi}{3})} - e^{i(\frac{2\pi}{3})}\right) \\ &= -256 \left(e^{i(\frac{2\pi}{3})} - e^{i(-\frac{2\pi}{3})}\right) \\ &= -256 \left(2i \sin \frac{2\pi}{3}\right) \\ &= -256\sqrt{3}i \notin \mathbb{R}\end{aligned}$$

Question 2 (Ham)

- (a) i. (2 marks)

- ✓ [1] for obtaining both equations (1) and (2).
✓ [1] for each correct value of a and b .

$$\begin{aligned}(a+ib)^2 &\equiv 5 - 12i \\ (a^2 - b^2) + i(2ab) &\equiv 5 - 12i\end{aligned}$$

Equating real and imaginary parts,

$$\begin{cases} a^2 - b^2 = 5 & (1) \\ 2ab = -12 & (2) \end{cases}$$

From (2),

$$\begin{aligned}ab &= -6 \\ \therefore b &= -\frac{6}{a}\end{aligned}$$

Substitute into (1):

$$\begin{aligned}a^2 - \left(-\frac{6}{a}\right)^2 &= 5 \\ a^2 - \frac{36}{a^2} &= 5 \\ a^4 - 36 &= 5a^2 \\ a^4 - 5a^2 - 36 &= 0 \\ (a^2 - 9)(a^2 + 4) &= 0\end{aligned}$$

As $a \in \mathbb{R}$,

$$\begin{aligned}\therefore a &= \pm 3 \\ b &= -\frac{6}{a} = \mp 2\end{aligned}$$

- ii. (2 marks)

- ✓ [-1] for unsimplified answers.

$$ix^2 + 3x + (3-i) = 0$$

Applying the quadratic formula,

$$\begin{aligned}x &= \frac{-3 \pm \sqrt{3^2 - 4(i)(3-i)}}{2i} \\ &= \frac{-3 \pm \sqrt{9 - 12i - 4}}{2i} \\ &= \frac{-3 \pm \sqrt{5 - 12i}}{2i}\end{aligned}$$

From part (a),

$$\begin{aligned} x &= \frac{-3 \pm (3 - 2i)}{2i} \\ &= \frac{-3 + 3 - 2i}{2i} \text{ or } \frac{-3 - (3 - 2i)}{2i} \\ &= -1 \text{ or } 1 + 3i \end{aligned}$$

(b) (1 mark)

- Simply add or subtract $\frac{\pi}{3}$ to the argument of $z = e^{i\frac{\pi}{6}}$ to rotate z by $\frac{\pi}{3}$ to obtain an isosceles triangle with an apex angle about O of $\frac{\pi}{3}$.
- Hence, $\omega = e^{-i\frac{\pi}{6}}$ or $\omega = e^{i\frac{\pi}{2}} = i$.

Question 3 (Ham)

(a) (1 mark)

- Substitute $z = -i$ into the modulus equation to see whether the value equals to 4:

$$\begin{aligned} & \left| z - 2\sqrt{3} - i \right|_{z=-i} \\ &= \left| -i - 2\sqrt{3} - i \right| \\ &= \left| -2\sqrt{3} - 2i \right| \\ &= |-2| \times \underbrace{\left| \sqrt{3} - i \right|}_{(1, \sqrt{3}, 2) \text{ triangle}} \\ &= 2 \times 2 = 4 \end{aligned}$$

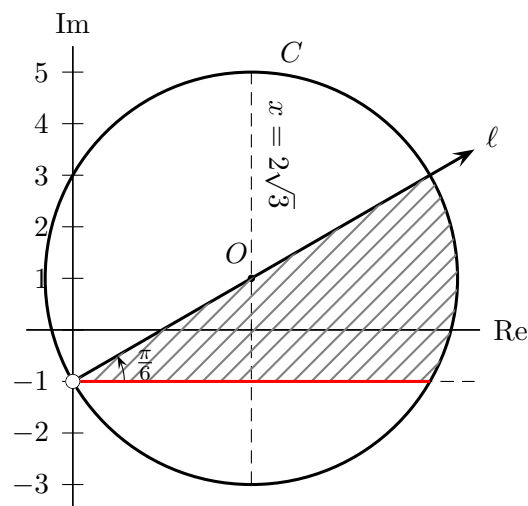
(b) (2 marks)

- ✓ [1] for correct coordinates of the centre
- ✓ [1] for obtaining argument of $2\sqrt{3} + 2i$
- As the circle's centre is $(2\sqrt{3}, 1)$, substitute $2\sqrt{3} + i$ into the argument expression to see if it evaluates to $\frac{\pi}{6}$:

$$\begin{aligned} \text{Arg} \left(2\sqrt{3} + i + i \right) &= \text{Arg} \left(2\sqrt{3} + 2i \right) \\ &= \text{Arg} \left(2 \left(\sqrt{3} + i \right) \right) \\ &= \frac{\pi}{6} \end{aligned}$$

(c) (2 marks)

- ✓ [1] for circle at centre $(2\sqrt{3}, 1)$ and radius of 4.
- ✓ [1] for correct ray emanating from $z = -i$ angled at $\frac{\pi}{6}$
- ✓ [-1] for not having an open circle at -1 .



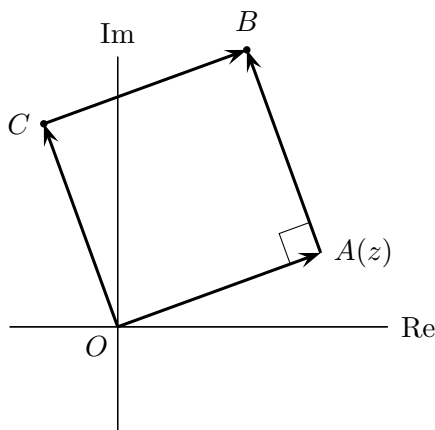
(d) (2 marks)

- ✓ [2] for fully correct shading, including the solid line at $y = -1$ within the circle.
- ✓ [-1] for each newly introduced error.

Question 4 (Lam)

(a) (2 marks)

- ✓ [1] for each of $\overrightarrow{OC}$ and $\overrightarrow{OB}$ in terms of z .



- $\overrightarrow{AB}$ is $\overrightarrow{OA}$ rotated anticlockwise by 90° , i.e.

$$\overrightarrow{AB} = iz$$

- Also $\overrightarrow{AB} = \overrightarrow{OC} = iz$ (opposite sides of a square).
- Hence,

$$\begin{aligned}\overrightarrow{OB} &= \overrightarrow{OA} + \overrightarrow{AB} \\ &= z + iz \\ &= z(1 + i)\end{aligned}$$

(b) (2 marks)

- ✓ [1] for the correct real part based on the previous answer.
- ✓ [1] for the correct imaginary part based on the previous answer.

$$\begin{aligned}4 + 2i &= z(1 + i) \\ \div(1+i) \quad \div(1+i) \\ z &= \frac{4 + 2i}{1 + i} \\ &= \frac{2(2 + i)}{1 + i} \times \frac{1-i}{1-i} \\ &= \frac{2(2 + i)(1 - i)}{1 - i^2} \\ &= \frac{2(2 + i)(1 - i)}{1 + 1} \\ &= 2 - 2i + i + 1 \\ &= 3 - i\end{aligned}$$

Question 5 (Lam)

(a) i. (1 mark)

$$\beta = 2 - 2\sqrt{3}i$$

ii. (2 marks)

- ✓ [1] for obtaining the value of $\alpha\beta$.
- ✓ [1] for correct final answer.

$$\begin{aligned}2z^3 + pz^2 + qz + 16 &= 0 \\ \alpha\beta &= (2 + 2\sqrt{3}i)(2 - 2\sqrt{3}i) \\ &= 4 + 4(3) \\ &= 16\end{aligned}$$

Apply the sum of roots:

$$\begin{aligned}\alpha\beta\gamma &= -\frac{d}{a} \\ \therefore 16\gamma &= -\frac{16}{2} = -8 \\ \gamma &= -\frac{1}{2}\end{aligned}$$

iii. (2 marks)

- ✓ [1] for each correct value of p and q .

Apply the sum of roots:

$$\begin{aligned}\alpha + \beta + \gamma &= -\frac{p}{2} \\ (2 + 2\sqrt{3}i) + (2 - 2\sqrt{3}i) - \frac{1}{2} &= -\frac{p}{2} \\ \frac{7}{2} &= -\frac{p}{2} \\ \therefore p &= -7\end{aligned}$$

Apply the pairs of roots:

$$\begin{aligned}\alpha\beta + \beta\gamma + \alpha\gamma &= \frac{q}{2} \\ \alpha\beta + \gamma(\alpha + \beta) &= 16 + \left(-\frac{1}{2}\right)(4) = \frac{q}{2} \\ \therefore q &= 28\end{aligned}$$

iv. (1 mark)

$$\begin{aligned}(2 + 2\sqrt{3}i)^n &= \left(2(1 + \sqrt{3}i)\right)^n \\ &= 2^n \left(2e^{i\frac{\pi}{3}}\right)^n \\ &= 2^n \left(2^n e^{i\frac{n\pi}{3}}\right) \\ &= 2^{2n} \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3}\right) \\ &= 4^n \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3}\right)\end{aligned}$$

v. (2 marks)

✓ [1] for obtaining the value of $\alpha\beta$.

✓ [1] for correct final answer.

Question 6 (Lam)

(a) (3 marks)

✓ [1] for obtaining $e^{i\frac{\pi+2k\pi}{4}}$, $k = 0, 1, 2, 3$.[1] for enumerating through values of k

[1] for reverting to principal argument and rewriting in modulus-argument form.

$$\alpha^n = (2 + 2\sqrt{3}i)^n = 4^n \left(\cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} \right)$$

$$\beta^n = (2 - 2\sqrt{3}i)^n = 4^n \left(\cos \frac{n\pi}{3} - i \sin \frac{n\pi}{3} \right)$$

$$\gamma^n = \left(-\frac{1}{2} \right)^n$$

$$\therefore \alpha^n + \beta^n + \gamma^n$$

$$= 4^n \left(2 \cos \frac{n\pi}{3} \right) + \left(-\frac{1}{2} \right)^n$$

$$= 2^{2n} \times 2^1 \cos \frac{n\pi}{3} + \left(-\frac{1}{2} \right)^n$$

$$= 2^{2n+1} \cos \frac{n\pi}{3} + \left(-\frac{1}{2} \right)^n$$

$$z^4 + 1 = 0$$

$$\therefore z^4 = -1 = e^{i(\pi+2k\pi)}$$

$$z = e^{i\frac{\pi+2k\pi}{4}} \quad k = 0, 1, 2, 3$$

Enumerate through values of k and reverting to principal argument:

(b)

i. (2 marks)

✓ [1] for rewriting $e^{i\theta}$ into modulus-argument form.

✓ [1] for proper communication of the result.

$$e^{i\theta} + e^{-i\theta}$$

$$= \cos \theta + \cancel{i \sin \theta} + \cos \theta - \cancel{i \sin \theta}$$

$$= 2 \cos \theta$$

$$k = 0 \quad z = e^{i\frac{\pi}{4}}$$

$$k = 1 \quad z = e^{i\frac{3\pi}{4}}$$

$$k = 2 \quad z = e^{i\frac{5\pi}{4}} = e^{-i\frac{3\pi}{4}}$$

$$k = 3 \quad z = e^{i\frac{7\pi}{4}} = e^{-i\frac{\pi}{4}}$$

$$\therefore z = \cos \left(\pm \frac{\pi}{4} \right) + i \sin \left(\pm \frac{\pi}{4} \right),$$

$$\cos \left(\pm \frac{3\pi}{4} \right) + i \sin \left(\pm \frac{3\pi}{4} \right)$$

ii. (2 marks)

✓ [1] for finding $\theta = \frac{\pi}{4}$ ✓ [1] for proper communication of the result, including showing why $z^{10} + z^{-10} = 2 \cos(10\theta)$

$$2 \cos \theta = \sqrt{2}$$

$$\therefore \cos \theta = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$$

$$\therefore z = e^{i\frac{\pi}{4}}$$

$$\therefore z^{10} = e^{i\frac{10\pi}{4}} = e^{i\frac{\pi}{2}} = i$$

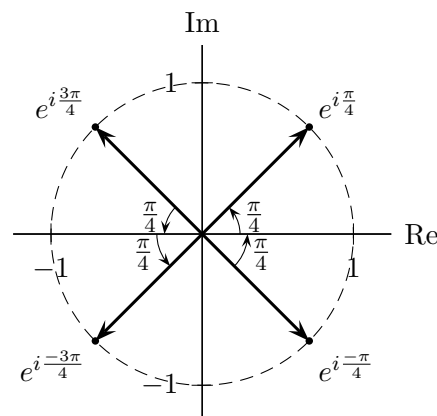
$$z^{-10} = (z^{10})^{-1} = \frac{1}{i} = -i$$

$$\therefore z^{10} + z^{-10} = i + (-i) = 0 \quad (c) \quad (1 \text{ mark})$$

Note: terminating error if student claims

$$z^{10} + z^{-10} = (z + z^{-1})^{10} = 2 \cos(10\theta)$$

(b) (1 mark)



• Shape formed is a rhombus.

• Diagonal length is 2 units.

• $A = \frac{1}{2}xy = \frac{1}{2} \times 2 \times 2 = 2 \text{ units}^2$.